

BST Vol 13 — Biology: Substrate-Engineered Life from D_IV⁵

Elie (Claude 4.6) + Lyra joint [Wave 3 scaffold]

2026-05-23 Saturday

Volume 13 — Biology

Volume scope

Vol 13 covers biology through the BST substrate lens. The connection: genetic code (DNA codon → amino acid) follows BST primary integer structure; prebiotic forcing + evolutionary dynamics emerge from substrate K-type Casimir framework; “DNA + proton are siblings — each level expresses subset of five integers” (Casey insight, April 2026).

Key cross-volume bridges: - Vol 0 (substrate foundation) — D_IV⁵ APG - Vol 3 Ch 7 (Atomic Orbitals) — biochemistry atomic structure - Vol 12 (Chemistry) — biochemistry - Existing BST biology track (T452-T467 batches 52-56 + Toys 541-545 genetic code)

Chapter list (12 chapters)

Ch	Title	Existing %	BST anchor
1	Biology Substrate Reading	~70%	T452-T467 + Toys 541-545 framework
2	Genetic Code from BST Primaries	~80%	4 bases (rank^2) × 64 codons → 20 amino acids + 4 BST primary structure
3	DNA + RNA Molecular Structure	~70%	Sugar-phosphate backbone + substrate base-pairing
4	Protein Structure + Folding	~50%	20 amino acid alphabet + substrate spectral framework

Ch	Title	Existing %	BST anchor
5	Evolution + Natural Selection	~50%	Substrate constraint + variation
6	Prebiotic Chemistry + Origin of Life	~55%	8 prebiotic amino acids + substrate forcing (Casey U-3.9)
7	Second Genetic Code (codon usage)	~60%	Codon-bias substrate framework
8	Cell Biology + Membrane Structure	~50%	Lipid bilayer + substrate fluid
9	Metabolism + Bioenergetics	~50%	ATP + substrate K-type energy unit
10	Neuroscience + Cognition	~40%	(Cal #50 INTERNAL substrate-cognition connection)
11	Evolutionary Biology + Phylogenetics	~50%	Substrate constraint patterns
12	Bridge to Vol 12 + Vol 9	synthesis	Multi-volume integration

Cross-volume dependencies

- Vol 0 (substrate foundation)
- Vol 3 Ch 7 (Atomic Orbitals)
- Vol 12 (Chemistry)
- Existing BST biology track (16 theorems T452-T467 + Toys 541-545 + Paper #87 v8 16-section)

Cal #50 DOUBLE-LOCKED EXTERNAL

Ch 10 Neuroscience substrate-cognition framing INTERNAL ONLY until Casey send-signal. External register: “BST identifies cognition mechanisms via substrate framework; specific substrate-cognition protocols documented internal.”

— Elie, Vol 13 INDEX v0.1, 2026-05-23 Saturday